

Open Source CFD

Tutorial #6: 3D Y-Pipe and Turbulence Models

Incompressible Steady Flow: 3D Y-Pipe with different turbulence models

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Learning Objectives:

- Switch between turbulence models in CfdOF
- Understand the y^+ parameter and its importance
- Compute and visualize y^+ directly in ParaView
- Compare laminar, $k-\varepsilon$, $k-\omega$ SST, and SSTLM models
- Determine if mesh is appropriate for each model using y^+
- Analyze how turbulence affects flow separation and mixing

This tutorial explores turbulence modeling using the Y-pipe geometry from Tutorial #5. Instead of editing dictionaries manually, you will use CfdOF's intuitive GUI to switch between turbulence models. You will learn how the y^+ parameter determines whether your mesh is suitable for each model, and how to visualize y^+ in ParaView. By the end, you'll understand why the same mesh can be perfect for one model and completely inadequate for another.

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1 Introduction to Turbulence Modeling in CfdOF

1.1 Why Turbulence Models Matter

Most engineering flows are turbulent. The choice of turbulence model affects:

- **Predicted pressure drop** (important for pump sizing)
- **Flow separation** (recirculation zones, mixing)
- **Mesh requirements** (y^+ constraints determine cell size near walls)
- **Computational cost** (more equations = slower)

1.2 Turbulence Models in CfdOF

CfdOF provides easy access to these RANS models:

Model	CfdOF menu option	Equations	y^+ requirement
Laminar	"Laminar"	0	N/A
Standard k - ε	"RANS k-epsilon"	2	> 30 (wall functions)
k - ω SST	"RANS k-omega SST"	2	< 1 or > 30
SSTLM	"RANS k-omega SSTLM"	4	< 1 (must resolve transition)

Table 1: Turbulence models available in CfdOF

1.3 The Critical Concept: y^+

y^+ is the dimensionless wall distance:

$$y^+ = \frac{yu_\tau}{\nu} \quad (1)$$

where $u_\tau = \sqrt{\tau_w/\rho}$ is the friction velocity.

The dimensionless velocity u^+ is function of the dimensionless distance to wall. For a turbulent boundary layer, it is assumed, and experimentally validated that the function is linear,

$$u^+ = y^+ \quad (2)$$

close to the wall ($y^+ \lesssim 5$), and it follows a logarithm law,

$$u^+ = B + \frac{1}{\kappa} \log(y^+) \quad (3)$$

, where $B \approx 5$ and $\kappa \approx 0.42$, at a distance $y^+ \gtrapprox 30$ (see Figure ??). For high Reynolds number, mesh the domain near the wall to get small values of y^+ can be very expensive in term of computational resources and time (very large mesh). This computation is avoided with the use of a **Wall Function** that models the behavior of the flow as far as y^+ is large enough, in the log-law region. The buffer region, $5 \lesssim y^+ \lesssim 30$ should be, in general, avoided.

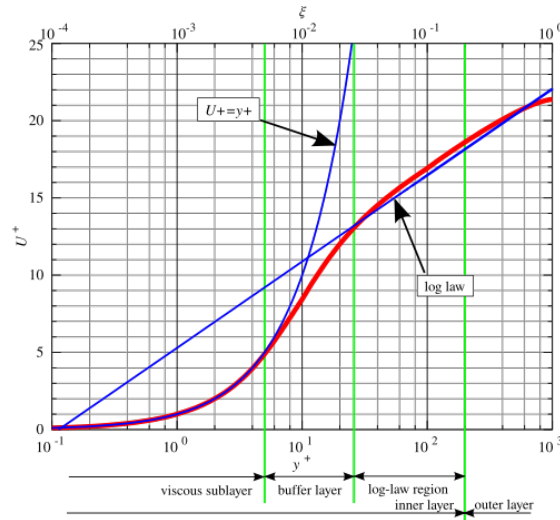


Figure 1: Law of the wall showing y^+ regions. From [Wikipedia](#)

2 Computation of y^+ for the $k - \omega$ SST simulation

In tutorial #5 we have performed the simulation with the $k - \omega$ SST turbulence model. Now we are going to perform the rest of simulations (Laminar, $k - \varepsilon$, $k - \omega$ SSTLM) in the same document. But, first, let's compute the value of y^+ for the simulation already done.

2.1 Setup of the FreeCAD-Cfd0f session

- Load the FreeCAD project from Tutorial #5 and save it as **Tutorial_6** in the proper folder.
- Change the **default output directory** in **Cfd0f preferences** to the Tutorial #6 folder
- Let's we tell **FreeCAD** to allow duplicate names in the tree. Open **Edit** → **Preferences** and, in **General/Document** check on the box for **Allow duplicate object labels** in one document. Close with OK

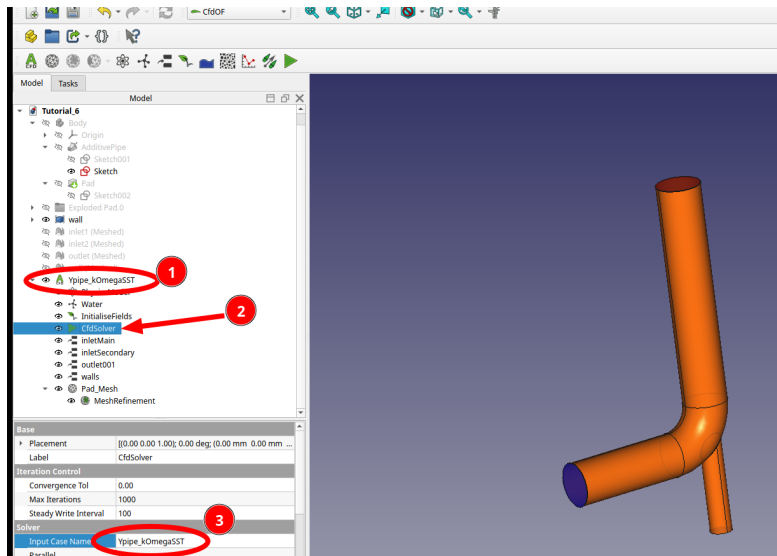


Figure 2: Changing names in the initial simulation

2.2 Making again the $k - \omega$ SST simulation

- Rename the CfdAnalysis as Ypipe_kOmegaSST
- Write and Run the case so that we will have it available in our new folder for Tutorial #6.
- The simulation can be stopped in 500 iterations, since the result will be almost the definitive one.

2.3 Estimation of y^+ with CLI

- In the Analysis control window click on Edit to open the file File Manager and with the right-button in the mouse select Open in Terminal. Alternatively, a terminal can be open with `ctrl-alt-t` and then navigate to the simulation's case. You should be in

```
eseiaat@eseiaat-linux:~/Tutorials/Tutorial_6/Ypipe_kOmegaSST$
```

- Load OpenFOAM

```
source /usr/lib/openfoam/openfoam2312/etc/bashrc
```

TIP: Remember the `tab` autocompletion.

- Get the dictionary to compute y^+ .

```
foamGetDict yPlus
```

- Run the yPlus function

```
simpleFoam -postProcess -func yPlus -latestTime
```

You will get the main information of the computation, similar to the output of Listing 1, and also a new file yPlus will be in the folder for the latest simulation time, e.g. 500.

```
Create time

Create mesh for time = 500


SIMPLE: convergence criteria
field "(p|p_rgh)" tolerance 0.001
field U tolerance 0.001
field "(k|epsilon|omega|f|v2|nuTilda|gammaInt|ReThetat)"
    tolerance 0.001


Time = 500
Reading field p

Reading field U

Reading/calculating face flux field phi


Selecting incompressible transport model Newtonian
Selecting turbulence model type RAS
Selecting RAS turbulence model kOmegaSST
Selecting patchDistMethod meshWave
RAS
{
    RASModel          kOmegaSST;
    turbulence         on;
    printCoeffs        on;
    alphaK1            0.85;
    alphaK2            1;
    alphaOmega1        0.5;
    alphaOmega2        0.856;
    gamma1             0.5555555555555556;
    gamma2             0.44;
    beta1              0.075;
    beta2              0.0828;
    betaStar           0.09;
    a1                 0.31;
    b1                 1;
    c1                 10;
```

```

F3                false;
decayControl      false;
kInf              0;
omegaInf          0;
}

No MRF models present

No finite volume options present
yPlus yPlus write:
writing field yPlus
patch walls y+ : min = 1.91730240538628, max =
                258.340024080099, average = 41.7649846515448

End

```

Listing 1: Output of yPlus function

TIP: The yPlus function, which calculates y^+ at the domain wall, supports many options. In this tutorial, the default settings are used, but you can view all of them on openfoam.com

- Click on **ParaView** in **FreeCAD** to open the results for the last time. Choose in the **OpenFOAMReader1** element just the **wall** patch
- Apply the **Threshold Filter** to show only the values of y^+ between 5 and 30. The size of the wall with these values is very large, as shown in Figure 3. This could be the reason of the poor convergence of the simulation,

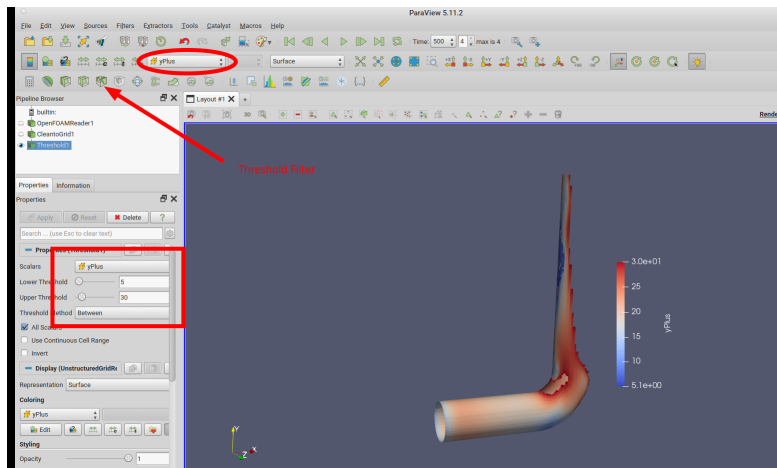


Figure 3: Region of the wall with y^+ between 5 and 30.

Note that the values of y^+ are very large (typically $\gtrsim 20$), so we should significantly refine the mesh close to the wall to get values smaller than 5. It would make the

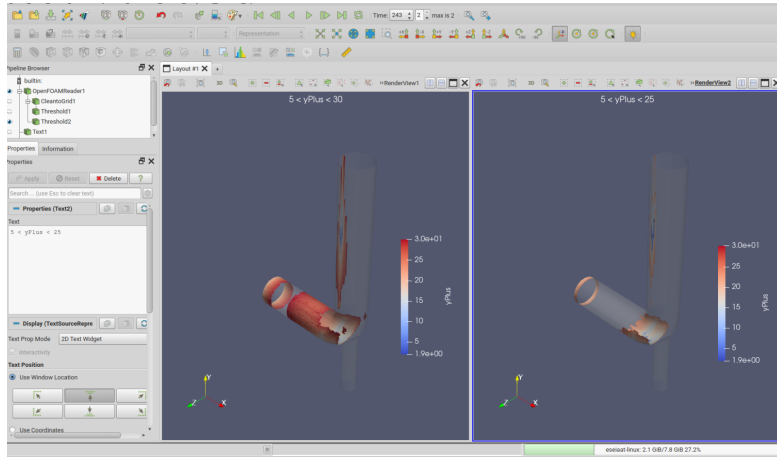


Figure 4: Distribution of y^+ with new mesh

simulation very slow. Alternatively, we are going to increase the size of the mesh near the wall to approach the log-law region.

2.4 Modification of mesh to improve value of y^+

- In the FreeCAD application, double-click in the Mesh refinement region, and change the number of layer in the boundary from 3 to 1.
- Write and Run the mesh.
- Write and Run the solver. Now the simulation converges much better. The values of y^+ are much larger, and the region where y^+ falls in the buffer zone is smaller. Figure 4 shows the threshold regions for $5 < y^+ < 30$ and $5 < y^+ < 25$

3 Laminar simulation

- Select the Ypipe_kOmegaSST element and click on the menu Edit → Duplicate selection. Leave selected all the dependences. A new Cfd Analysis container is created in the tree. Change its name to Ypipe_Laminar.
- It has to be activated with a double-click.
- In the CfdSolver of this new container, change the Input case name to Ypipe_Laminar
- The case has to be remeshed.
- Write and Run the solver. Note that a new case named Ypipe_Laminar is created in the Tutorial_6 folder

- The simulation is fast but the convergence is very bad, because the lack of a turbulence model.

4 $k - \varepsilon$ simulation

- Repeat the same steps than in laminar simulation but with $k - \varepsilon$ model. Change the name according to this.
- This simulation will need more iterations. Change the maximum number of iterations to 2000. It will take about 1000 iterations to converge.

5 Note on SSTLM (Langtry-Menter 4-Equation Model)

The SSTLM model is designed to predict laminar-to-turbulent transition and requires:

- $y^+ < 1$ at wall-adjacent cells
- **Much finer mesh** (typically $10-50\times$ more cells near walls)
- **Significantly longer** computation time (hours instead of minutes)

IMPORTANT: With our current mesh ($y^+ \gtrapprox 30$), SSTLM will produce incorrect results because it cannot resolve the laminar sublayer where transition occurs. For this reason, we will not run SSTLM in this tutorial, but you should be aware of its existence and requirements for when you encounter transitional flows (e.g., low Reynolds number airfoils, turbomachinery blades).

6 Exercises

6.1 Flow Rate Comparison

Using ParaView's **Integrate Variables** filter on each simulation:

1. Open each case in ParaView (or load them together for comparison)
2. Select the **inletSecondary** patch
3. Apply **Integrate Variables** filter
4. Record the volumetric flow rate $Q = \int U \cdot dA$
5. Create a table like Table 2:
6. Which model predicts the highest flow rate? The lowest? Why?

Model	Flow rate (m ³ /s)	Iterations to converge	Notes
Laminar			Unstable convergence
k - ε			Good convergence
k - ω SST			Very good convergence

Table 2: Flow rate comparison across turbulence models

6.2 Pressure Drop: Main Inlet to Outlet

Although both have zero-gradient or fixed pressure, you can still calculate:

$$\Delta p_{main-outlet} = p_{main,inlet} - p_{outlet}$$

Since $p_{outlet} = 0$ (fixed), this becomes simply $p_{main,inlet}$.

1. In ParaView, select the `inletMain` patch.
2. Compute the average pressure value (it will not be zero!)
3. Compare across turbulence models:

Model	$p_{main,inlet}$ (Pa)
Laminar	
k - ε	
k - ω SST	

Table 3: Pressure at main inlet for different models

4. Which model requires the highest pressure to drive the flow? Why?